

Algebra 1
Unit 6
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

Alanna made a mistake when solving the inequality $\frac{-7n+15}{8} - 5 \geq 3$. For questions 1-4, use Alanna's work below.

Given Inequality	$\frac{-7n+15}{8} - 5 \geq 3$
Step 1	$-7n + 15 - 5 \geq 24$
Step 2	$-7n + 10 \geq 24$
Step 3	$-7n \geq 14$
Step 4	$n \leq -2$

1. Identify the step in which Alanna made an error. Write a note to Alanna explaining why that step was incorrect.

Alanna made an error in Step 1.

Alanna, when "un-doing" the division of 8, you must multiply every term in the inequality by 8. You forgot to multiply the (-5) by 8.

2. Solve the inequality $\frac{-7n+15}{8} - 5 \geq 3$.

$$\begin{array}{l} \frac{-7n+15}{8} - 5 \geq 3 \\ \quad +5 \quad +5 \\ \left(\frac{8}{1}\right) \frac{-7n+15}{8} \geq 8 \left(\frac{8}{1}\right) \\ -7n + 15 \geq 64 \\ \quad -15 \quad -15 \end{array}$$

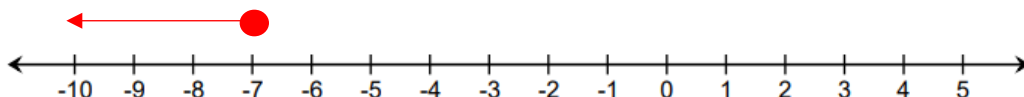
$$\frac{-7n}{-7} \geq \frac{49}{-7}$$

$n \leq -7$

3. Write the solution in set-builder notation.

$$\{n \in \mathbb{R}: n \leq -7\}$$

4. Graph the solution set.



Use the inequality $0.06m + 2.56 > -0.15m + 8.86$ to complete questions 5-7.

5. Solve the inequality.

$$\begin{array}{r} 0.06m + 2.56 > -0.15m + 8.86 \\ +0.15m \quad \quad +0.15m \end{array}$$

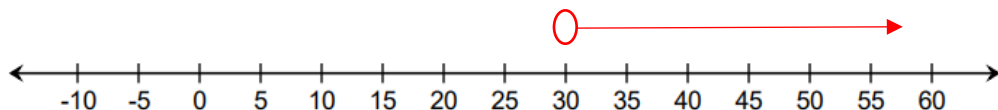
$$\begin{array}{r} 0.21m + 2.56 > 8.86 \\ -2.56 \quad -2.56 \end{array}$$

$$\frac{0.21m}{0.21} > \frac{6.3}{0.21} \quad m > 30$$

6. Write the solution in set-builder notation.

$$\{m \in \mathbb{R}: m > 30\}$$

7. Graph the solution set.



Use the inequality $-\frac{1}{2}(5n + 2) \leq -\frac{1}{3}(10n - 2)$ to complete questions 8-10

8. Solve the inequality.

$$\left(\frac{6}{1}\right)\left(-\frac{1}{2}(5n + 2)\right) \leq \left(\frac{6}{1}\right)\left(-\frac{1}{3}(10n - 2)\right)$$

$$-3(5n + 2) \leq -2(10n - 2)$$

$$\begin{array}{r} -15n - 6 \leq -20n + 4 \\ +20n \quad +20n \end{array}$$

$$5n - 6 \leq 4$$

$$\begin{array}{r} 5n - 6 \leq 4 \\ +6 \quad +6 \end{array}$$

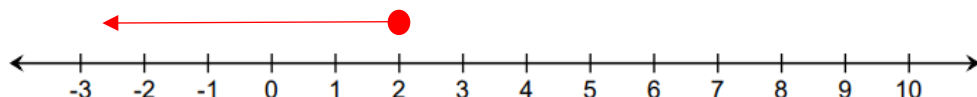
$$5n \leq 10$$

$$n \leq 2$$

9. Write the solution in set-builder notation.

$$\{n \in \mathbb{R}: n \leq 2\}$$

10. Graph the solution set.



Algebra 1
Unit 6
Lesson 2 Practice

Name Answer Key
Date _____
Period _____

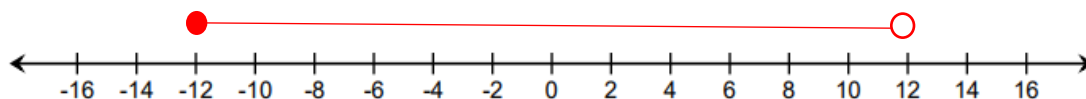
Use the following conjunctive inequality to answer questions 1-3.

$$-24 - 2x \leq -\frac{x}{3} - 4 < -2x + 16$$

Rewrite the compound inequality as two single inequalities and solve each one.

1. (3) (3) (3) (3) $-24 - 2x \leq -\frac{x}{3} - 4$ $-72 - 6x \leq -x - 12$ $+6x \quad +6x$ $-72 \leq 5x - 12$ $+12 \quad +12$ $\frac{-60}{5} \leq \frac{5x}{5} \quad -12 \leq x \quad \boxed{x \geq -12}$	2. (3) (3) (3) (3) $-\frac{x}{3} - 4 < -2x + 16$ $-x - 12 < -6x + 48$ $+6x \quad +6x$ $5x - 12 < 48$ $+12 \quad +12$ $\frac{5x}{5} < \frac{60}{5} \quad \boxed{x < 12}$
---	--

3. Graph the solution set of the compound inequality.



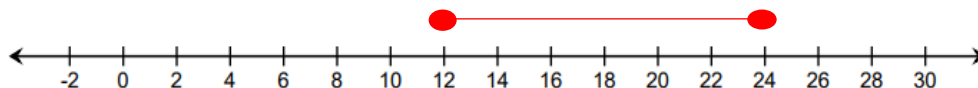
Alex just started a new sales floor job to save for college. He earns \$16.25 per hour plus a flat fee of \$55 per week. He wants to earn between \$250 and \$445 per week. The following inequality represents his earning potential:

$$250 \leq 16.25x + 55 \leq 445$$

4. Solve the inequality.

$$\begin{array}{lcl} 250 \leq 16.25x + 55 & \text{and} & 16.25x + 55 \leq 445 \\ -55 & & -55 \\ \hline \frac{195}{16.25} \leq \frac{16.25x}{16.25} & & \frac{16.25x}{16.25} \leq \frac{390}{16.25} \\ 12 \leq x & & x \leq 24 \end{array}$$

5. Graph the solution on the number line.



6. Complete the statement: Alex can work between 12 and 24 hours each week.

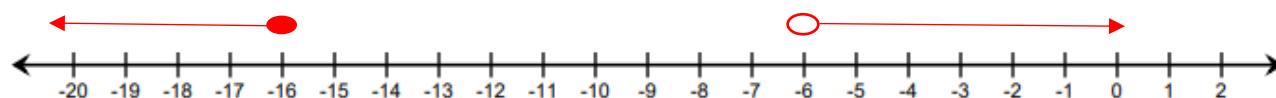
Use the disjunctive inequality shown below to answer questions 7-10:

$$-\frac{5}{3}\left(\frac{4}{3}k + \frac{8}{5}\right) > -3\frac{1}{4}k - \frac{53}{6} \text{ or } \frac{24k+14}{5} \leq -74$$

Solve each component of the compound inequality.

7. $-\frac{5}{3}\left(\frac{4}{3}k + \frac{8}{5}\right) > -3\frac{1}{4}k - \frac{53}{6}$ $\left(\frac{12}{1}\right)\left(-\frac{5}{3}\left(\frac{4}{3}k + \frac{8}{5}\right)\right) > \left(-\frac{13}{4}k - \frac{53}{6}\right)\left(\frac{12}{1}\right)$ $-20\left(\frac{4}{3}k + \frac{8}{5}\right) > -39k - 106$ $\frac{-80}{3}k - 32 > -39k - 106$ $+39k \quad +39k$ $\frac{37}{3}k - 32 > -106$ $+32 \quad +32$ $\left(\frac{3}{37}\right)\frac{37}{3}k > -74\left(\frac{3}{37}\right)$ $k > -6$	8. $\left(\frac{5}{1}\right)\left(\frac{24k+14}{5}\right) \leq -74\left(\frac{5}{1}\right)$ $24k + 14 \leq -370$ $-14 \quad -14$ $\frac{24k}{24} \leq \frac{-384}{24}$ $k \leq -16$
---	---

9. Graph the solution sets of both inequalities on the number line.



10. Explain which of the following values is not a solution to the inequality.

-19	-9	1
-----	----	---

-9 is not a solution to the inequality because it does not lie in the solution set.

In addition, -9 is not shaded on the graph of the number line.

Algebra 1
Unit 6
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

Amar is saving for a trip to Costa Rica with his friends. He has estimated that his travel expenses will cost between \$1,800 and \$2,400 total. He has \$300 in savings, and he earns \$16.50 per hour at his job. He wants to know how many hours he must work to afford the trip.

1. Write an inequality to represent the situation.

$$1800 \leq 300 + 16.50x \leq 2400$$

2. Solve the inequality.

$$\begin{array}{r} 1800 \leq 300 + 16.50x \\ -300 \quad -300 \end{array}$$

$$\frac{1500}{16.50} \leq \frac{16.50x}{16.50}$$

$$90.91 \leq x$$

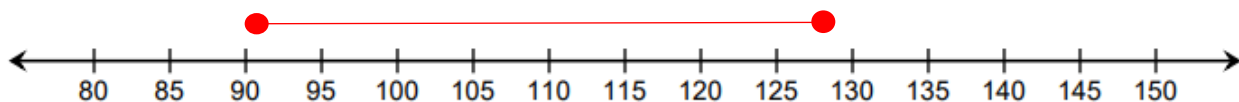
$$\begin{array}{r} 300 + 16.50x \leq 2400 \\ -300 \quad -300 \end{array}$$

$$\frac{16.50x}{16.50} \leq \frac{2100}{16.50}$$

$$x \leq 127.27$$

He must work between 90.91 and 127.27 hours to afford his trip.

3. Graph the solution.



4. Check any that apply.

- ☒ Solutions must be non-negative.
☐ Solutions must be integers.

5. Explain your selection(s) from question 4.

He can work fractions of hours, but cannot work negative hours.

Marcel just started her own business making cookies. She sells the cookies in boxes, charging \$15.00 per box. It costs her \$50.00 to buy the supplies. She wants to calculate how many boxes she must make to earn a profit of at least \$245.00.

6. Write an inequality to represent the situation.

$$15x - 50 \geq 245$$

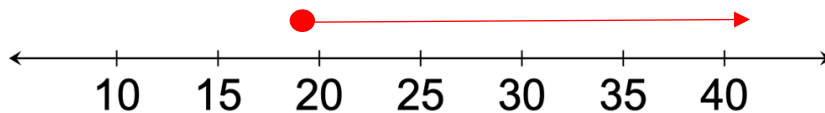
7. Solve the inequality.
- $$\begin{array}{r} 15x - 50 \geq 245 \\ +50 \quad +50 \end{array}$$

$$\frac{15x}{15} \geq \frac{295}{15}$$

$$x \geq 19.\bar{6}$$

She would need to sell at least 20 boxes to make a profit of at least \$245.00.

8. Graph the solution.



9. Check any that apply.

- ☒ Solutions must be non-negative.
☒ Solutions must be integers.

10. Explain your selection(s) from question 9.

She needs to sell whole boxes not fractions of boxes and she cannot sell negative boxes.

Algebra 1
Unit 6
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

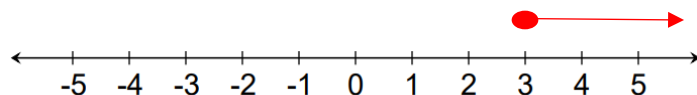
Consider the inequality $-3x + 4 \leq -5$.

1. Solve the inequality. $-3x + 4 \leq -5$
 $\quad\quad\quad -4 \quad -4$

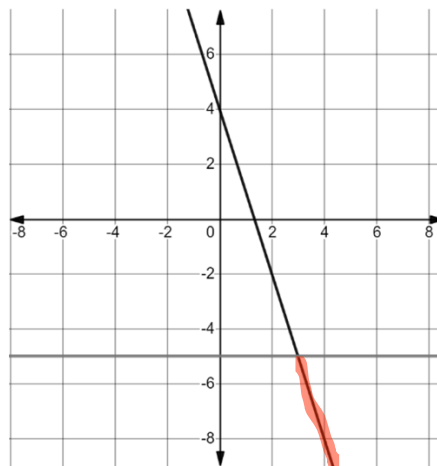
$$\frac{-3x}{-3} \leq \frac{-9}{-3}$$

$$x \geq 3$$

2. Graph the solution set.



3. The graphs of $y = -3x + 4$ and $y = -5$ are shown on the coordinate plane.

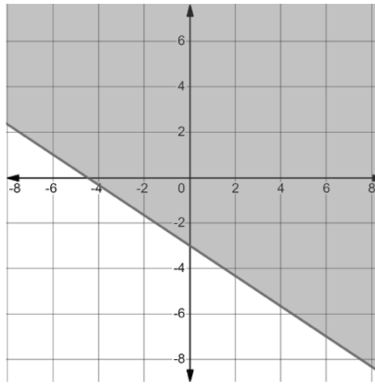


Highlight the part of the graph of $y = -3x + 4$, where the y values are less than or equal to -5 .

4. Describe the relationship between the solution to the inequality $-3x + 4 \leq -5$ and the values of x for which the graph of $y = -3x + 4$ is less than or equal to -5 .

When $x \geq 3$, the solution to $-3x + 4 \leq -5$, the y -values are less than or equal to -5 .

The graph of the two-variable inequality $y \geq -\frac{2}{3}x - 3$ is shown.



5. The slanted solid line is called the **boundary line** of the inequality. Write the equation of the boundary line in slope-intercept form.

$$m = \frac{-4}{6} = \frac{-2}{3} \quad y = \frac{-2}{3}x - 3$$

6. How does the equation of the boundary line relate to the inequality?

The equation of the boundary line is the same as the inequality except that it has an equal sign.

7. Describe what part(s) of the graph represent the solution set of the inequality $y \geq -\frac{2}{3}x - 3$.

All values on and above the boundary line are in the solution set to the inequality.

8. Which part of the graph of the solution set of $y \geq -\frac{2}{3}x - 3$ corresponds to each mathematical statement?

Mathematical Statement	Graphical Representation
$y > -\frac{2}{3}x - 3$	☐ boundary line ☒ shaded region
$y = -\frac{2}{3}x - 3$	☒ boundary line ☐ shaded region

Complete the statements.

9. If the points on the line are included in the solution set, the boundary line

is

- ☐ dotted.
☒ solid.

10. If the points on the line are not included in the solution set, the boundary line

is

- ☒ dotted.
☐ solid.

Algebra 1
Unit 6
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

The linear inequality $6x - 3y < 18$ is written in standard form.

1. Complete the statement.

The boundary line of this inequality is

- ☒ dotted
☐ solid

because the points on the

boundary line are

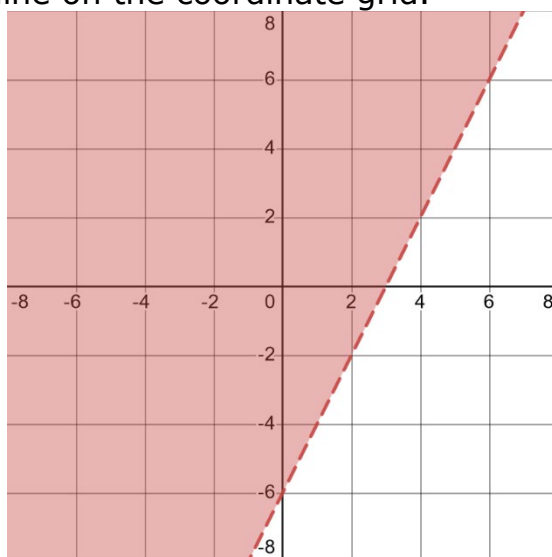
- ☐ included from
☒ excluded from

the set.

2. Write the equation of the boundary line of this inequality in standard form.

Answers may vary. $6x - 3y = 18$

3. Graph the boundary line on the coordinate grid.



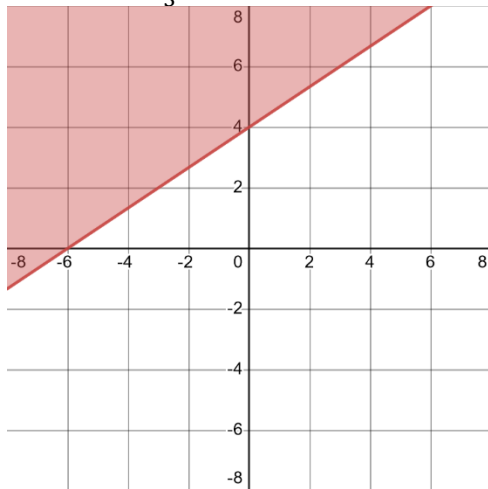
4. Explain how you will determine which side of the boundary line to shade.

Since the y -values must be greater than $2x - 6$, everything above the boundary line gets shaded since $y > 2x - 6$.

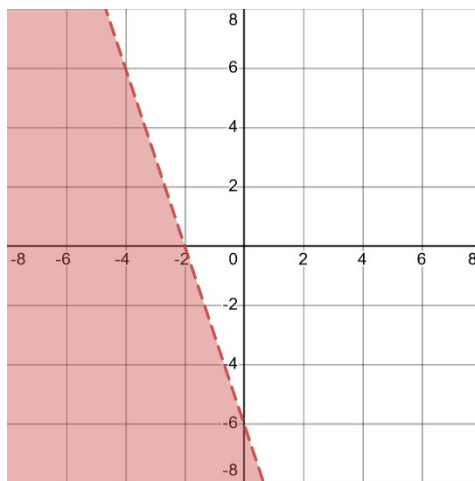
5. Shade the region on the graph that represents the solution set.

For each problem below, graph the given inequality.

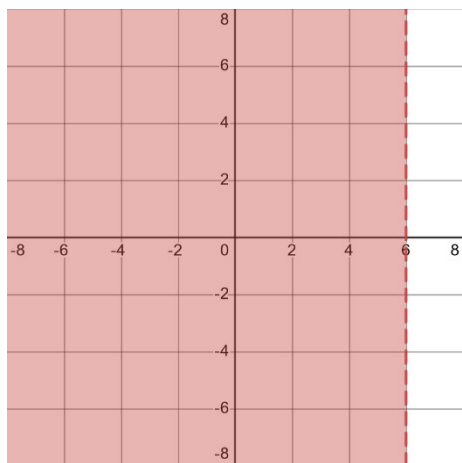
6. $y \geq \frac{2}{3}x + 4$



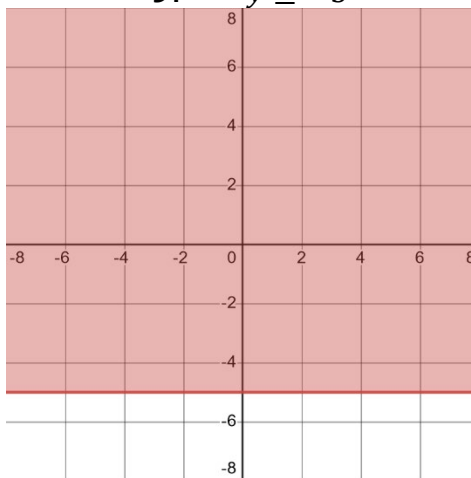
7. $y < -3x - 6$



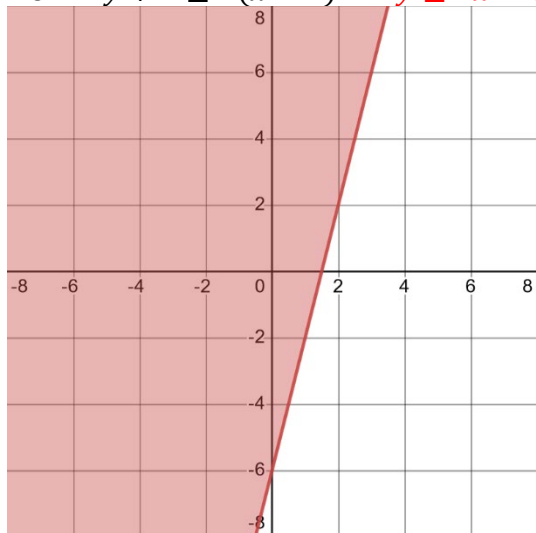
8. $x < 6$



9. $y \geq -5$



10. $y + 2 \geq 4(x - 1) \rightarrow y \geq 4x - 6$



Algebra 1
Unit 6
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

Jacqueline is buying grapes and apples from the Walmart produce section. Grapes are \$3 per pound and apples are \$4 per pound. Jacqueline's mother gave her \$20 to spend. To determine the combinations of grapes and apples that she can afford to purchase, Jacqueline wrote the inequality $3x + 4y \leq 20$.

1. Define the variables Jacqueline used based on the context.

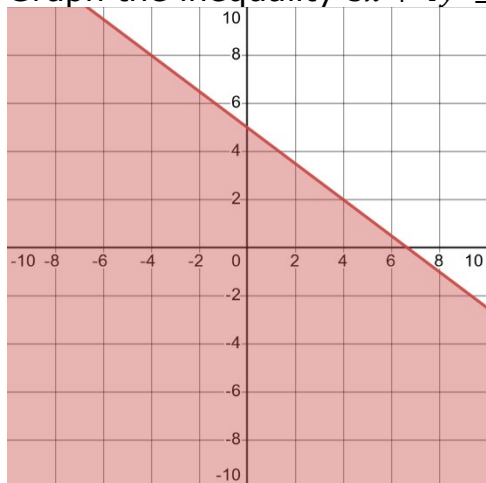
Let x represent the weight of grapes purchased, in pounds.

Let y represent the weight of purchased apples, in pounds.

2. Explain why the inequality sign $\leq$ makes sense in this context.

The inequality sign, $\leq$, in this context makes sense because it means they can spend up to their max, which is \$20.

3. Graph the inequality $3x + 4y \leq 20$.



4. The ordered pair $(3, -4)$ is a solution to the inequality $3x + 4y \leq 20$. Explain why this is not a viable solution in the context of buying grapes and apples.

It is not viable because y represents the weight of apples purchased, which cannot be negative.

5. The ordered pair $(4, 2)$ is a point on the boundary line. Explain what the ordered pair represents in the situation.

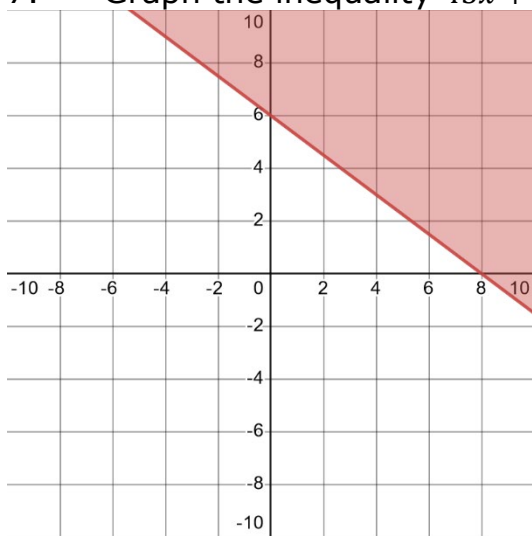
Jacqueline can purchase 4 pounds of grapes and 2 pounds of apples for exactly \$20.

Miguel invested some of his savings in the stock market. He bought 45 shares of a video production company and 60 shares of a clothing company. He hopes to make a combined profit of at least \$360 when he sells his shares. He wrote the inequality $45x + 60y \geq 360$ to represent the situation, where x is the change in value of one share of the video production company and y is the change in value of one share of the clothing company.

6. Explain what it would mean if x or y were a negative number.

The value of the shares will have decreased in value from the time of Miguel's initial investment.

7. Graph the inequality $45x + 60y \geq 360$.



8. Explain what the point $(-2.50, 9.75)$ represents in Miguel's situation.

Shares of the movie stock decreased by \$2.50, but shares of the clothing stock increased by \$19.75, resulting in a combined profit of $45(-2.5) + 60(9.75) = \$472.50$, which is ≥ 360 .

9. Explain why it makes sense in Miguel's context that there are no solutions for which both x and y are both negative.

If both x and y are negative, profit earned will never be positive, let alone equal to \$360.

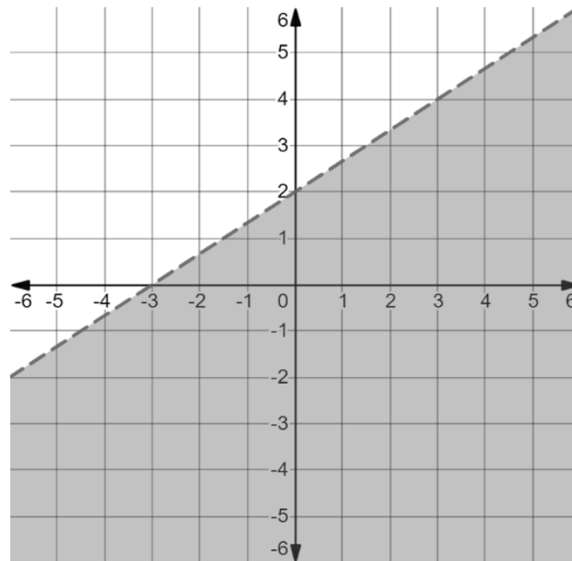
10. Stock prices in the United States are rounded to the nearest hundredths place. Complete the statement:

In order for solutions to the mathematical inequality to be viable solutions in the real-world context, the x and y values must be rounded to 2 decimal places.

Algebra 1
Unit 6
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

The graph of a two variable linear inequality is shown.



1. Write the inequality of the solution set graphed above in slope-intercept form.

$$y < \frac{2}{3}x + 2$$

2. Explain how to determine which inequality symbol to use when writing an inequality in slope-intercept form.

- The shading is below the line so when it is below the sign is $<$.
- You can also test out a point to check for accuracy.
- Since the boundary line is dotted, it must be $<$ or $>$.

3. Write the inequality of the solution set graphed above in point-slope form.

Answers may vary.

$$y - 4 < \frac{2}{3}(x - 3)$$

4. Explain how to determine which inequality symbol to use when writing an inequality in point-slope form.

Test a point or based on shading it is $<$ for below and $>$ for above.

5. Write the inequality of the solution set graphed above in standard form.

$$\begin{array}{l} y < \frac{2}{3}x + 2 \\ -2 \quad -2 \\ y - 2 < \frac{2}{3}x \\ -y \quad -y \end{array} \quad \begin{array}{l} -2 < \frac{2}{3}x - y \\ 3(-2 < \frac{2}{3}x - y) \\ -6 < 2x - 3y \\ = 2x - 3y > -6 \end{array}$$

Five inequalities are given. Complete the table below to match each inequality to the graph of its solution set by writing the inequality on the line provided under each graph.

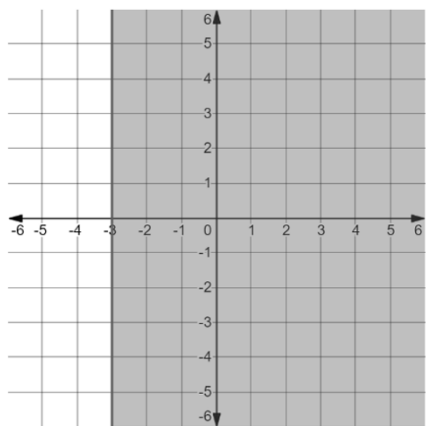
$$x \geq -3$$

$$y \leq -3x + 1$$

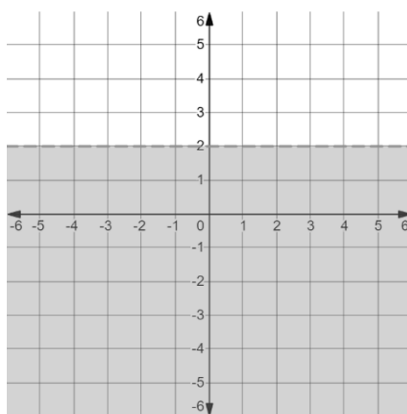
$$x - 2y > -6$$

$$y < 2$$

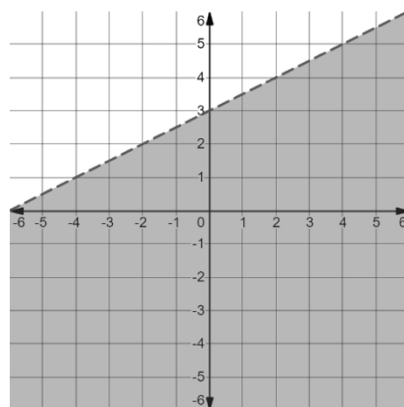
$$y + 1 > -\frac{3}{4}(x + 2)$$



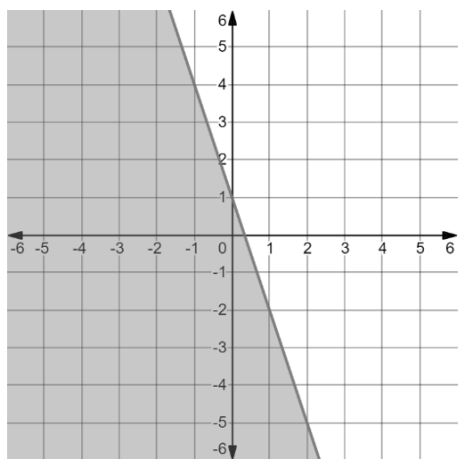
6. Inequality: $x \geq -3$



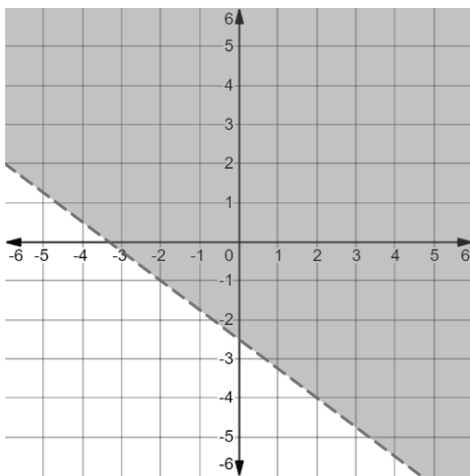
7. Inequality: $y < 2$



8. Inequality: $x - 2y > -6$



9. Inequality: $y \leq -3x + 1$



10. Inequality: $y + 1 > -\frac{3}{4}(x + 2)$

Algebra 1
Unit 6
Lesson 8 Practice

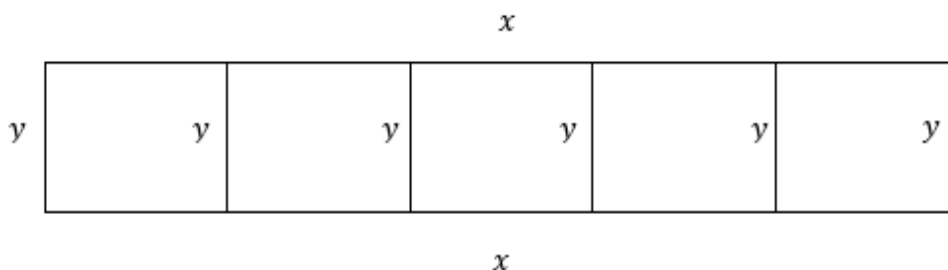
Name Answer Key
Date _____
Period _____

Jaime wants to build a rectangular fence in his yard that will be divided into five equal pens for chickens. Each pen needs to be the same size. He has 36 feet of fencing to build the fence. This scenario can be modeled as a two-variable inequality.

1. Describe what the two variables would represent in the context of Jaime's fence.

x will represent length of the rectangular fence.
 y is the sides of the rectangular fence and individual pens.

2. The diagram below shows a possible model of Jaime's fence with sides labeled as x and y .



3. Recall Jaime had 36 feet of fencing to use. Write a two-variable inequality that Jaime could use to find possible dimensions for his fence.

$$2x + 6y \leq 36$$

4. Explain how you chose which inequality symbol to use.

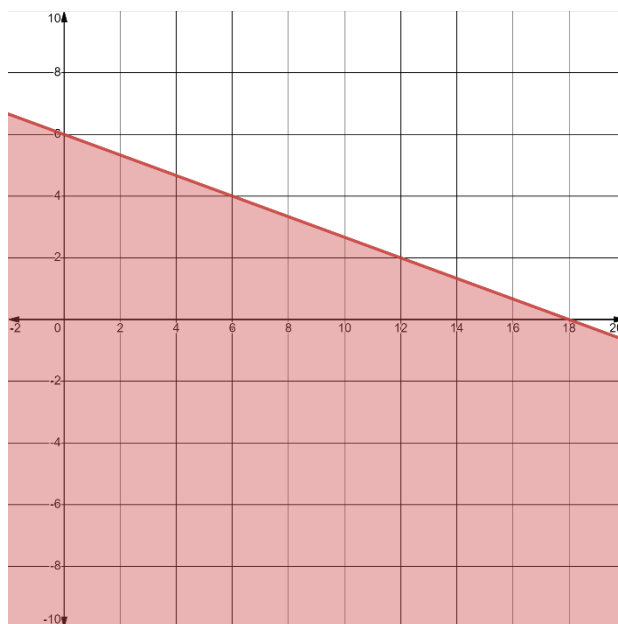
He only has 36 ft. so he can use at most 36 ft., so he needs to use less than or equal to 36.

5. Graph the inequality.

$$\begin{array}{rcl} 2x + 6y & \leq & 36 \\ -2x & & -2x \end{array}$$

$$\frac{6y}{6} \leq \frac{-2x+36}{6}$$

$$y \leq -\frac{1}{3}x + 6$$



Eric was applying for a personal loan to pay for home improvement projects. His bank requires a minimum of \$600 between his checking and savings accounts to be considered for a loan.

6. Define two variables for this situation.

Let x = checking account balance

Let y = savings account balance

7. Write an inequality in terms of x and y that could be used to determine whether Eric meets the minimum requirement for the loan.

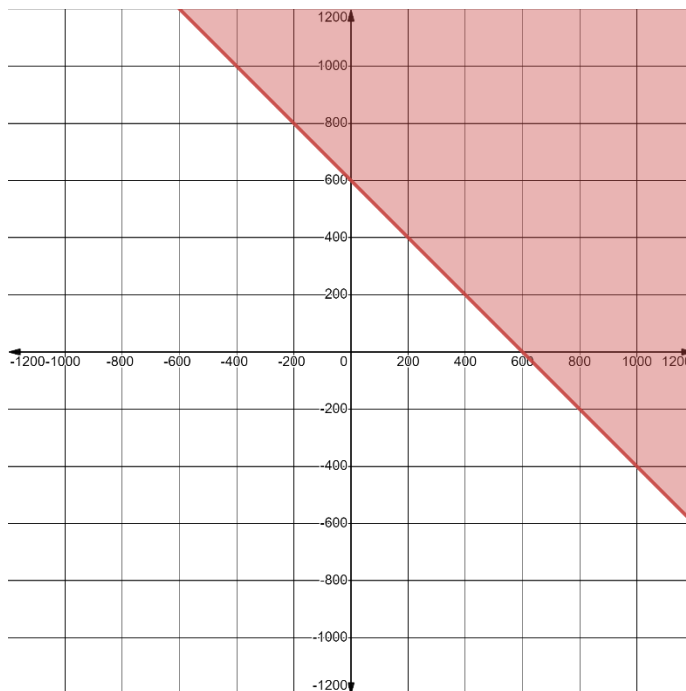
$$x + y \geq 600$$

8. Graph the inequality.

$$x + y \geq 600$$

$$-x \quad -x$$

$$y \geq -x + 600$$



9. Complete the statement.

To be a viable solution in this context, the x and y values must both be rounded to 2 decimal places. Money is rounded to 2 decimal places.

10. Eric overdrew his checking account, which now has a balance of $-\$275.00$. He has $\$957.23$ in his savings account. Does he meet the minimum requirement?

$$\begin{array}{r} 957.23 \\ -275.00 \\ \hline 682.23 \end{array}$$

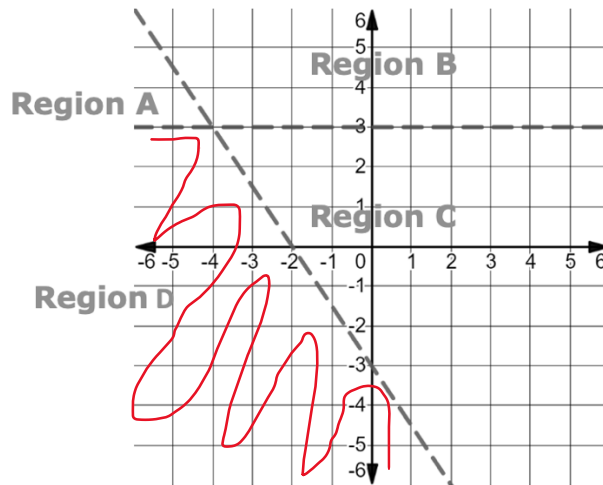
He has a positive balance of \$682.23 so he does have enough.

Algebra 1
Unit 6
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

Consider the system of inequalities: $\begin{cases} y < 3 \\ 6x + 4y < -12 \end{cases}$

The boundary lines of both inequalities in the system are shown on the coordinate plane.



1. The boundary lines separate the plane into four regions, labeled A – D. Which region includes the solution set of the system of inequalities?

Region D (shade below $y < 3$ and below $6x + 4y < -12$)

2. Shade the region that includes the solution set of the system of inequalities.

3. Select a test point in the shaded region of the solution set. $(-3, 0)$
Answers will vary but must be from Region D.
 $y < 3$
 $0 < 3$ ✓

4. Substitute the x –values and the y –values of the test point to show that the ordered pair satisfies both inequalities.

$y < 3$	$6x + 4y < -12$
$(-3, 0)$ $0 < 3$ ✓	$(-3, 0)$ $6(-3) + 4(0) < -12$ $-18 + 0 < -12$ $-18 < -12$ ✓

5. Your friend says that the point $(-4, -3)$ is a solution to the system of inequalities. Explain whether or not you agree.

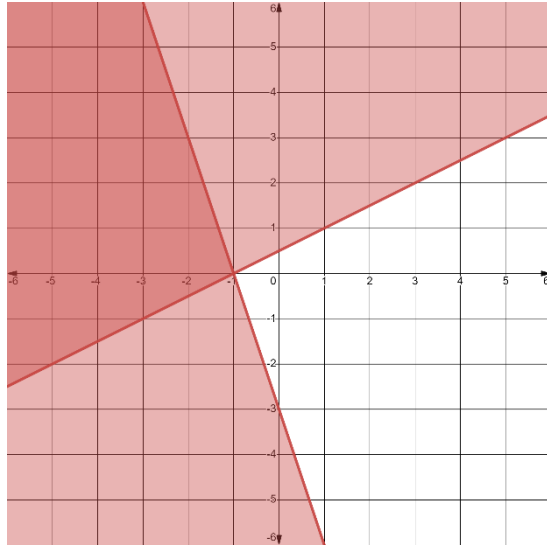
$(-4, -3)$
 $y < 3$
 $-3 < 3$ ✓

$6x + 4y < -12$
 $6(-4) + 4(-3) < -12$
 $-36 < -12$ ✓

The ordered pair makes both equations true, so I agree.

Consider the system of inequalities: $\begin{cases} y + 2 \geq 0.5(x + 5) \\ y - 3 \leq -3(x + 2) \end{cases}$

6. Graph the boundary lines of the system.

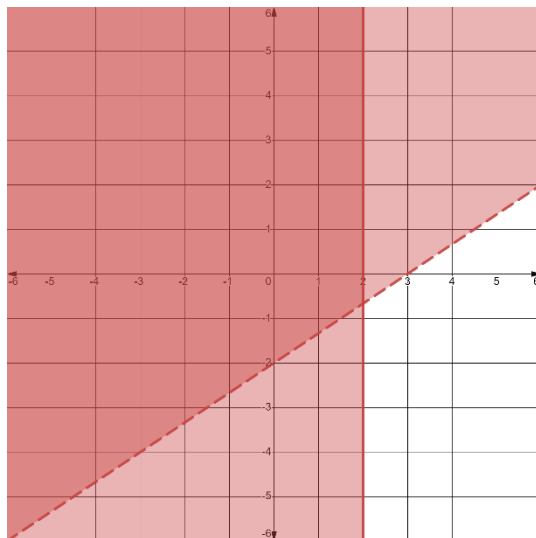


Use substitution to determine whether the given ordered pairs are solutions to the system.

	$y + 2 \geq 0.5(x + 5)$	$y - 3 < -3(x + 2)$	Solution to the system?
7. $(-3, 4)$	$4 + 2 \geq 0.5(-3 + 5)$ $6 \geq 1 \checkmark$	$4 - 3 \leq -3(-3 + 2)$ $1 \leq 3 \checkmark$	☒ Yes ☐ No
8. $(5, 2)$	$2 + 2 \geq 0.5(5 + 5)$ $4 \geq 5 \quad \times$	$2 - 3 \leq -3(5 + 2)$ $-1 \leq -21 \quad \times$	☐ Yes ☒ No

9. Complete the graph by shading the region included in the solution set of the system.

10. Graph the solution set of the system of inequalities: $\begin{cases} x \leq 2 \\ y > \frac{2}{3}x - 2 \end{cases}$

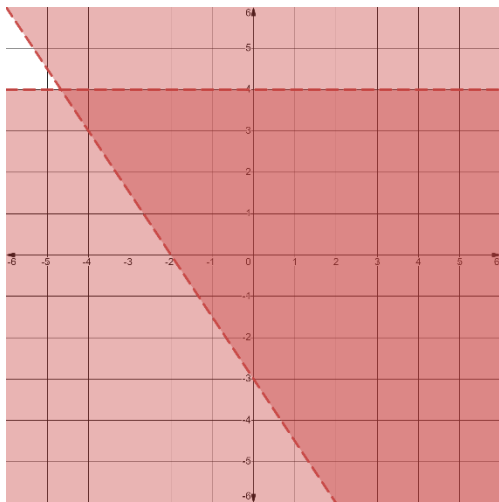


Algebra 1
Unit 6
Lesson 10 Practice

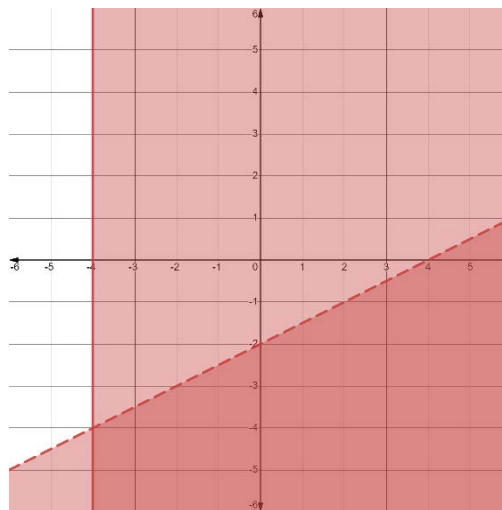
Name Answer Key
Date _____
Period _____

Graph the solution set for each system of inequalities on the coordinate grid.

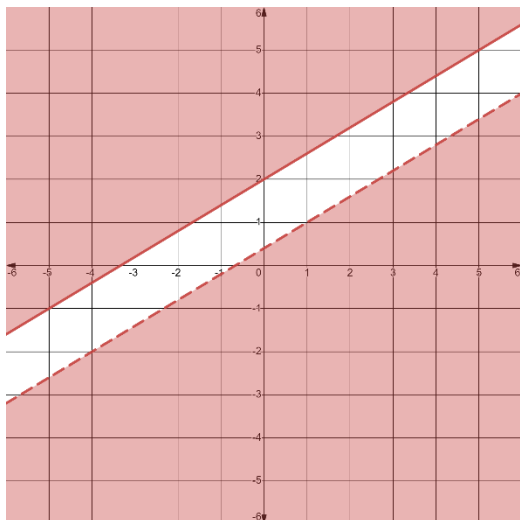
1.
$$\begin{cases} y < 4 \\ 3x + 2y > -6 \end{cases}$$



2.
$$\begin{cases} x \geq -4 \\ x - 2y > 4 \end{cases}$$



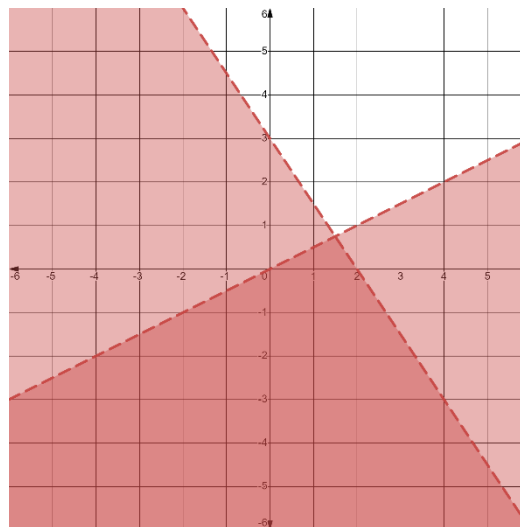
3.
$$\begin{cases} y \geq 0.6x + 2 \\ y - 1 < 0.6(x - 1) \end{cases}$$



$$y - 1 < \frac{3}{5}x - 0.6$$

$$y < \frac{3}{5}x + 0.4$$

4.
$$\begin{cases} y < -\frac{3}{2}x + 3 \\ y - 2 < 0.5(x - 4) \end{cases}$$

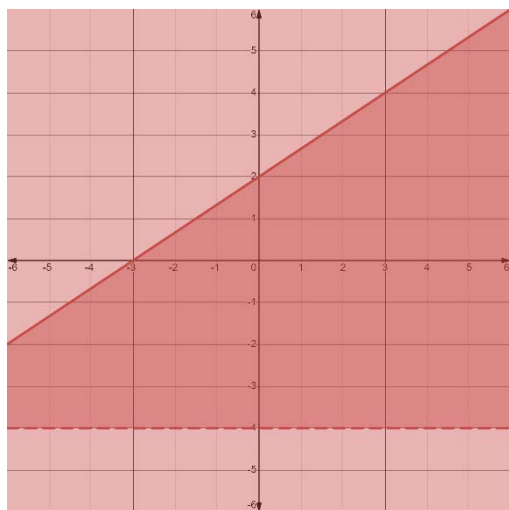


$$y - 2 < 0.5(x - 4)$$

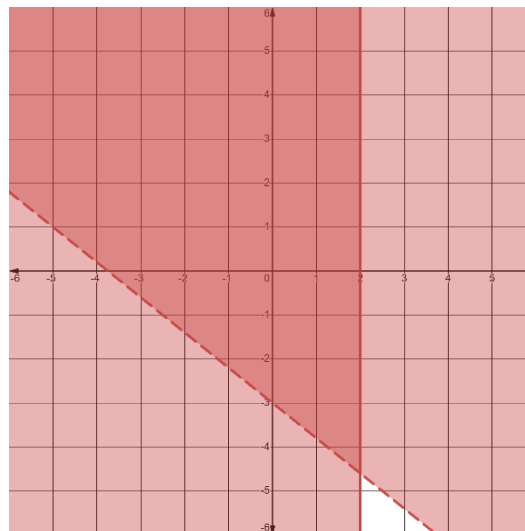
$$y - 2 < \frac{1}{2}x - 2$$

$$y < \frac{1}{2}x$$

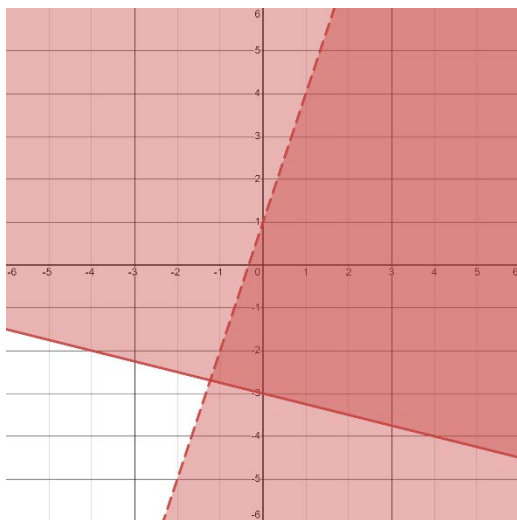
5.
$$\begin{cases} y > -4 \\ y + 2 \leq \frac{2}{3}(x + 6) \end{cases}$$



6.
$$\begin{cases} x \leq 2 \\ y > -\frac{4}{5}x - 3 \end{cases}$$



7.
$$\begin{cases} y \geq -\frac{1}{4}x - 3 \\ y < 3x + 1 \end{cases}$$



8.
$$\begin{cases} x \geq -3 \\ y > -5 \end{cases}$$

